Henry Hsu

Houston, TX 77025

+1-510-996-6837 www.yarikama.com yarikama yarikama hsuhengjui@gmail.com

SUMMARY

AI engineer working on both sides of agent systems: shipping them, and attacking them. Reproduced a tool-schema poisoning attack on Llama 3.1-8B and built a representation-level defense that cut privacy leakage 14.5pp with no loss in benign task success. Previously GenAI Team Lead at MaiAgent, where production AI agents with tool APIs and MCP grew users from 3K to 20K and cut token usage 67%. 14 merged PRs to LlamaIndex. M.C.S. at Rice, graduating Dec. 2026, available Feb. 2027.

WORK EXPERIENCE

Software Engineering Intern — Jun. 2026 – Aug. 2026

Google — *Taipei, Taiwan*

- **Integration Test Infrastructure:** Built **3** test infrastructures from a **zero** baseline (Java, Python), onboarding **400+** integration test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Agentic Failure Triage:** Developed an LLM agent that triages scheduled integration test runs and surfaces root-cause summaries directly on issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Device Reproduction:** Extended the agent to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Generative AI Team Lead — Dec. 2024 – Aug. 2025

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Building AI Agents:** Upgraded chatbots into AI agents by integrating memory systems, tool APIs, and MCP Client, resulting in **120%** partner growth (CTBC Bank, MSI, HPE, iGroup) and user expansion from **3K** to **20K** users while cutting LLM token usage by **67%+**
- **Contributing to Open Source:** Contributed **14** merged pull requests to the **LlamaIndex** open-source project (**15K+ stars**), fixing bugs and adding features in integrations with AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, MCP Client, and AI Agent Workflow
- **Testing Framework:** Introduced pytest testing framework for unit and E2E coverage to prevent production regressions, achieving **67%** test coverage from **zero** baseline, which reduced production hotfixes by **90%** initially and sustaining a **50%** reduction long term
- **Technical Roadmap & Leadership:** Defined GenAI product roadmap for a 10-person startup; led feasibility evaluation and end-to-end delivery of production features including Agentic RAG, Artifact Generation, Tagging & Permission System, and Information Retrieval

Generative AI Intern — Sep. 2024 – Dec. 2024

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Pipeline Scaling:** Refactored with async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks while **doubling** indexing performance
- **Reranker Integration:** Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5** and **7% Precision@10**

EDUCATION

Rice University - Top 20 U.S. University — Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00) — *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University — Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) — *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Defending LLM Agents Against Tool Schema Poisoning | *PyTorch, Llama 3.1, Activation Steering, Slurm* — **Apr. 2026 – May 2026**

- **Threat Model:** Reproduced a tool-schema poisoning attack where deceptive parameter descriptions make structured-output agents auto-fill private data, setting a **56%** attack-success and **52%** privacy-leak baseline on Llama 3.1-8B
- **Representation-Level Defense:** Probed hidden states to locate a linear safety direction (**96%** probe accuracy at layer 10) and steered activations along it at inference, cutting privacy leakage **14.5pp** with no loss in benign task success
- **Defense Comparison:** Ran a 2×2 single/multi-turn factorial showing the in-context reflection baseline buys its leak reduction with a **6pp** higher attack rate and **6pp** lower task success, both of which steering avoids

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Urban noise agent: structured-output LLM pipeline and tool calling

23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, Java, C/C++, TypeScript | **GenAI:** LlamaIndex, LangChain, LangGraph, MCP | **Backend:** Django, FastAPI
AI Security: Activation Steering, Hidden-State Probing, Red-Team Evaluators, Structured-Output Attack Surfaces, Tool-Call Guardrails
ML: PyTorch, numpy, pandas, scikit-learn | **Data:** PostgreSQL, Milvus, Elasticsearch | **Infra:** Docker, git, Slurm, AWS Bedrock/EC2/S3